

OPTICAL MEMS INDEX FINGER MICROGESTURE INPUT SENSOR FOR MOBILE AND WEARABLE DEVICES

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ABSTRACT

We report the first optical microsystem that captures index finger microgestures with the precision, bandwidth, power consumption and form-factor required for close-range handwriting and gesture keyboarding (e.g. Swipe) on smart watches and mobile handsets. Finger position is measured with 30 μ m resolution (at a 5cm distance) at a bandwidth of up to 550Hz while consuming less than 15mW with the reported prototype system. A VCSEL, scanning diffractive optic element (DOE), and photodiode may be packaged in a 2x2x1mm package to fit unobtrusively in a wearable device. This component may enable the first information-rich touch-less gesture inputs to wearables.

KEYWORDS

Gesture input, microgesture, human-computer interaction

INTRODUCTION

Background

At around the age of 1, human infants learn to point the index finger in order to direct attention and communicate relatively complex thoughts [1]. In most people, the index finger is the most dexterous digit in the hand and is capable of the finest motor skills. The proliferation of capacitive touch interfaces may be attributed to the intuitive and precise nature of finger gestures; however, this paradigm constrains human-computer interaction (HCI) to 2D surfaces. LIDAR modules [2] have extended HCI to 3D volumes in a number of consumer products, and ultrasonic gesture recognition [3] holds great promise as a low-power alternative. Index finger microgesture recognition for smart watches and mobile handsets requires high speed and resolution at short distances (1-10cm); it is in this niche that the reported system achieves unprecedented specifications.

Operating Principle

The present system uses a CMOS-MEMS scanning DOE to collimate and sweep the beam from an infrared (IR) VCSEL over a hemispheric region that is projected from a compact module (Fig. 1). When the IR beam is incident on the user's index finger, light is scattered and reflected. At the 850nm wavelength of the IR source, the specular component of reflection dominates over the Lambertian component, as scattering is proportional to $1/(\lambda^4)$. A photodiode captures light that is reflected back from the index finger. The detector is collocated with the DOE near the center of the hemisphere so that the mirror angle may be mapped to the position of the finger in

spherical coordinates. This configuration provides the highest resolution on finger movements in the center of the tracking range. The system reported in [4] relies on the geometry of the eye and its circular trajectory; the present system may be used to track objects with arbitrary trajectories that direct light back towards the detector. The components of the system may be integrated in a compact module to fit into wearable devices.

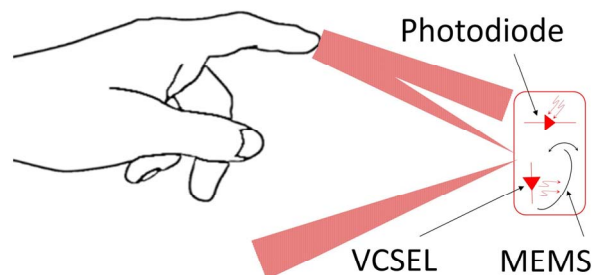


Figure 1: Cartoon depiction of the finger tracking sensor. A MEMS DOE scans the beam from a VCSEL over a region to illuminate the finger. The reflected beam from the finger is captured by a photodiode.

PROTOTYPE MODULE

A prototype module was assembled by hand to obtain experimental results (Fig.2). The scanner module consists of a lensed VCSEL that is packaged in a "pill-pack," a scanning MEMS DOE, and a photodiode. The scanner is positioned at an offset angle with respect to the photodiode plane in order to align the center of the scan range with the normal to the photodiode surface.

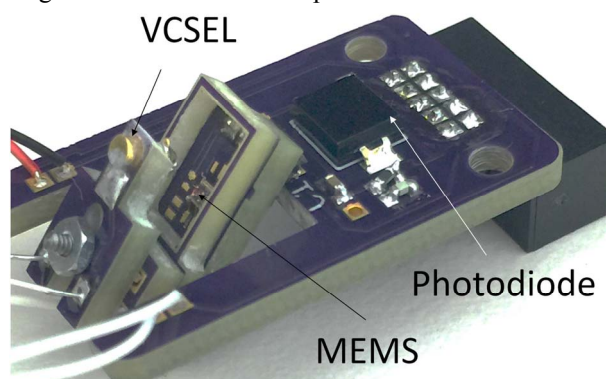


Figure 2: Hand-assembled prototype scanner module includes a lensed VCSEL in a "pill-pack," the MEMS scanner, and a photodiode with an IR filter.

The MEMS scanner device employs a serpentine design pattern of alternating bimorphs and rigid beams to achieve large angular deflections (Fig.3). The fast axis is operated in resonance with a scan range of 75-degrees

(mechanical), while the slow axis is operated in a quasi-static manner to achieve an 80-degree mechanical scan range. The slow axis is operated isothermally [4] in order to suppress temperature changes that may shift the natural frequency of the fast axis.

The payload of the scanner is a reflective Fresnel zone plate that simultaneously collimates and redirects the beam. This DOE enables the use of a lens-less scanner module that consists of a bare-die VCSEL that is placed at the focus of the DOE (Fig. 4). The $\sim 4.5 \times 6.5 \times 1.5$ mm ceramic package houses the MEMS scanner while the glass lid is patterned with traces to power the VCSEL. The waist of the beam projected by the scanner is larger than the VCSEL die, reducing the extent of the “blind-spot” in the scan range created by the VCSEL’s shadow.

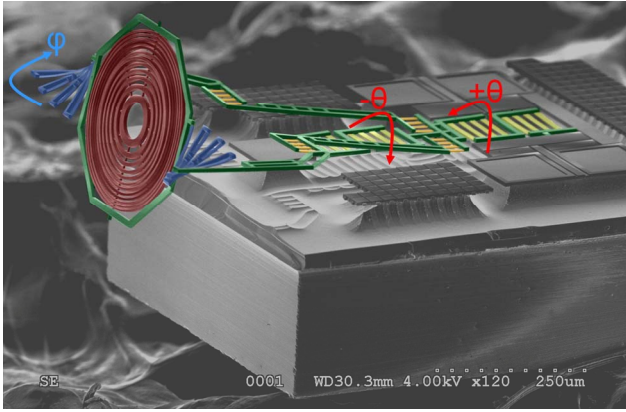


Figure 3: SEM image of the MEMS DOE scanner. The fast axis (blue, ϕ) is operated in resonance. Opposed serpentine actuators are operated isothermally to achieve angular deflections in the slow axis (yellow, θ). The DOE is suspended over the die edge in order to increase its range of motion.

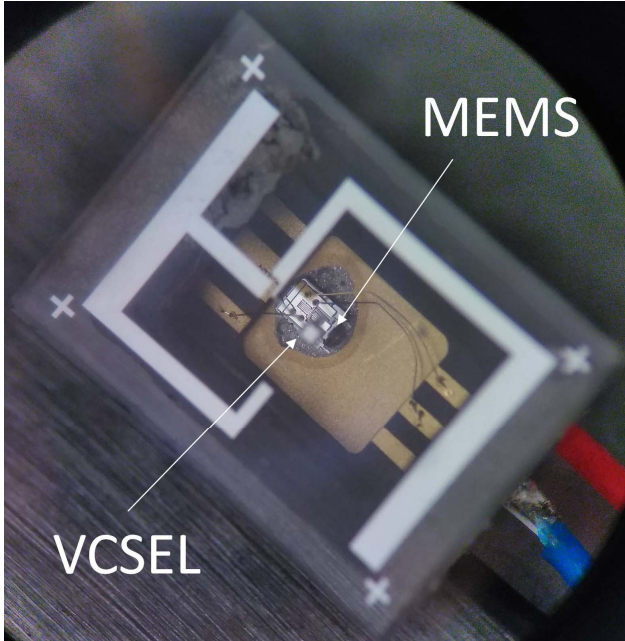


Figure 4: Compact scanner module before alignment step. The VCSEL is located on the glass cap with patterned signal routing and the MEMS scanner is die-bonded to a ceramic package. Package dimensions are $\sim 4.5 \times 6.5 \times 1.5$ mm.

TRACKING ALGORITHM

The prototype module in Figure 2 was used to obtain experimental results. The MEMS device was operated at resonance (fast axis: 1.5kHz-3.3kHz) to obtain pulses from the finger that are translated to position using an edge-triggered timer. Two pulses (forward and reverse) are obtained in each cycle (Fig. 5). The amplitudes of the peaks are captured by a peak-hold circuit that is employed for slow-axis tracking.

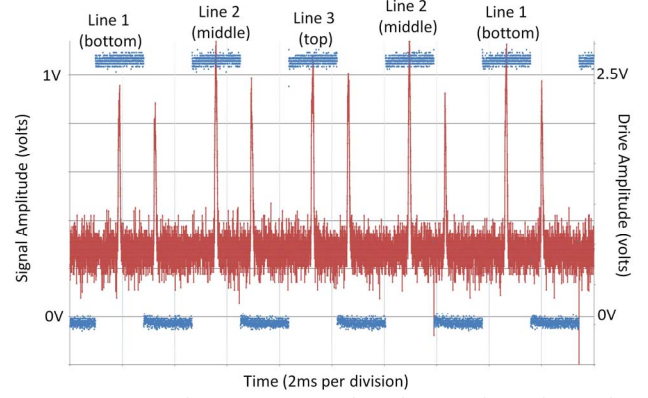


Figure 5: 2 peaks are captured within each cycle of the resonant axis, driven by a square wave.

An edge-tracking algorithm is also used to measure position in the slow (y-direction) axis (Fig.6). The y-axis position of the scanner is rastered over three levels such that the difference between the peak amplitudes above (a, a') and below (c, c') the centerline (b, b') are held constant. The peak reflected intensity is used to measure distance (z-direction).

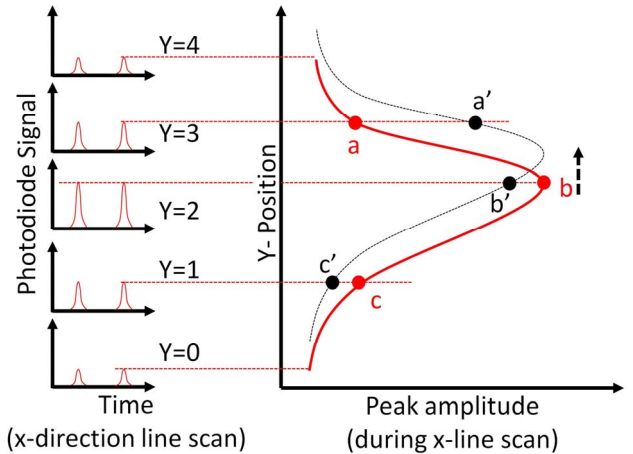


Figure 6: Vertical tracking algorithm employs the peak-held values in each line scan to track the global maximum region

EXPERIMENTAL RESULTS

To confirm that the scanner module is capable of capturing the position of an index finger in the x, y, and z directions, a user was instructed to wiggle their finger in the x and y directions, then to approach and retract from the module (Fig. 7). The inter-axis coupling evident in the data is due to the disparity between the spherical coordinate system of the sensor and the Cartesian coordinate system of the finger translations.

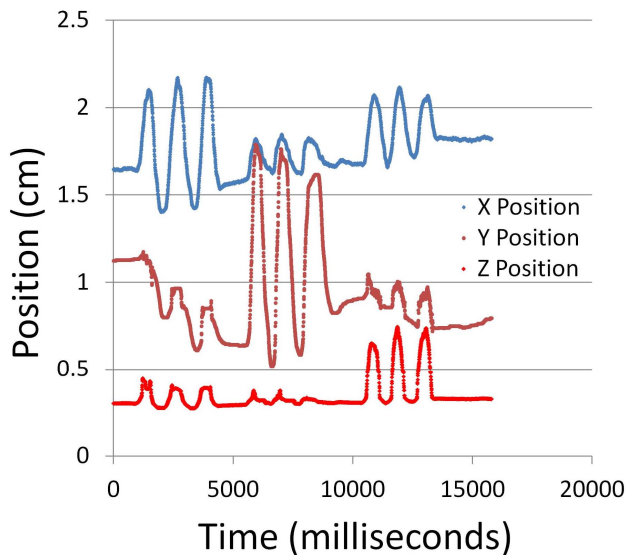


Figure 7: Data captured while a user was instructed to move their index finger in the x , y , and z directions sequentially

The signal from a 1cm deflection was compared to the noise measured while tracking a stationary object to obtain a resolution of $30\mu\text{m}$. The unpackaged MEMS device is susceptible to $1/f$ noise from ambient fluctuations. The combination of a packaged device and modulation techniques will further improve the resolution. The average tremor in the index finger is $30\mu\text{m}$, implying that this system may capture the tiniest voluntary movements.

Handwriting with small gestures was performed with the reported device. A user was instructed to write "trans2017" with the index finger to produce the plot in Figure 8. When gesturing over small volumes, the index finger may be stabilized using the thumb in order to suppress tremors that may obscure the intended movements.

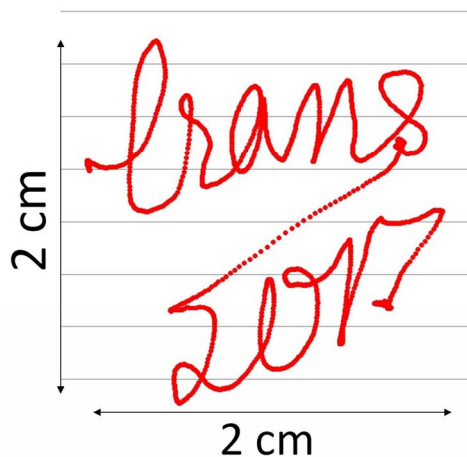


Figure 8: Data captured while a user wrote the words "trans 2017."

3D gestures were captured in the 10cm radius of the tracking hemisphere. A 3cm long helix with a 2cm diameter was drawn and is plotted in Figure 9.

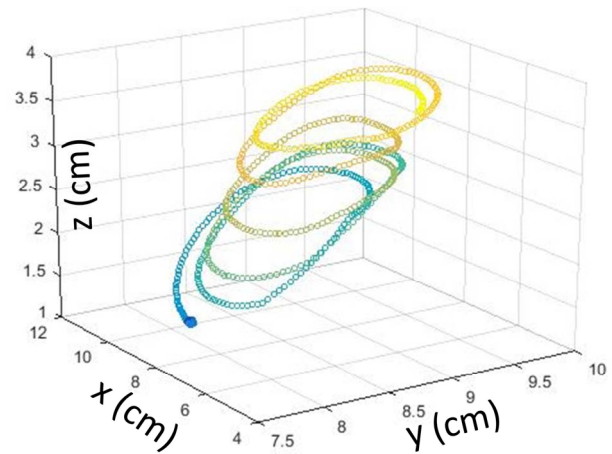


Figure 9: 3D gesture captured by the present tracking system. 5 turns of a 3cm long helix pattern with a 2cm diameter were drawn.

A star, circle, square and triangle were drawn at different depths to produce the plot in Figure 10.

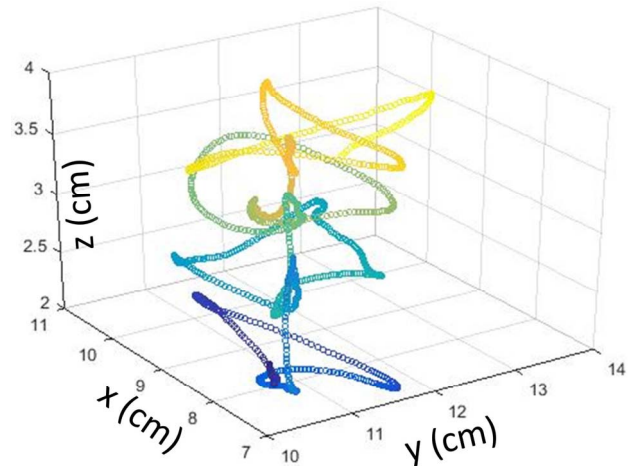


Figure 10: Complex gestural inputs may be captured with the prototype tracking system. Unfiltered data corresponds to a user drawing various shapes at different distances from the module.

SUMMARY

A compact scanner module is desirable for applications in wearable consumer products like smart-watches and fitness trackers. The components of the reported module are shown in Figure 11, along with a depiction of their intended integration into a watch. A number of compelling applications may be enabled by the present 3D tracking device.

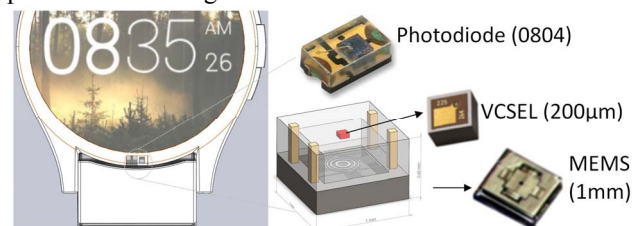


Figure 11: Depiction of a compact scanner module integrated in a smart-watch. The size of the photodiode, VCSEL, and MEMS scanner are shown separately (not to scale).

The reported system enables the interpretation of complex gestures like those required for Swipe keyboards, web-browsing, and menu navigation. Fast, precise gesture recognition in smart watches and other wearables (glasses, rings, bands) may significantly augment their utility.

ACKNOWLEDGEMENTS

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